Modelling

Higher · Design · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ explain how models are used to generate and explore, test and refine, and communicate
- ☐ describe the purpose of, and information gained from, test rigs and prototypes
- ☐ explain the use of computer models and simulations
- ☐ describe the benefits and drawbacks of rapid prototyping

I DO — Worked example

Read how a full-mark answer is built

1 Describe how physical models can be used during the design of a product. (4 marks)

Full-mark answer each ✓ is one mark

Sketch models made quickly from card let the designer try out several forms and check proportions early on ✓. A block model can be held to test the grip and button positions against hand sizes ✓. A test model or rig can check strength and stability, so faults are found before tooling is made ✓. A scale model shows the size of a large product relative to a person ✓.

Examiner tip: 2025 commentaries credited four distinct purposes; naming types alone and CAD models scored 0.

Why answers lose marks

"Sketch models, block models, prototypes and CAD models."

0 marks: naming model types scores nothing, and computer models are not physical models.

WE DO — Guided practice

Use the starters and word bank to help you

2 Discuss how designers use a variety of models during the design process. Link each to the stage, purpose and information gained. (8 marks)

Sentence starters

Scale models: …

Computer models and simulations: …

Word bank

sketch · generate · explore · models · ergonomics · client · proportion · refine

Tip: Marked in bands: cover several stages, each with an example and why it is used.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) An architect shows a new school building to the council. Which model?
☐ Sketch model ☐ Scale model ☐ Test rig ☐ Prototype

b) Which is a benefit of simulation?
☐ It uses the real material ☐ Testing without risk to users or the environment ☐ It needs no computer ☐ It is always cheaper than sketching

c) Why are sketch models used early in the design process?
☐ They show the final finish ☐ They are quick and cheap to make and change ☐ They are fully working ☐ They are used for mass production

d) Which model would a designer use to test the comfort of a handle grip?
☐ Scale model ☐ Block model ☐ Sketch model of a different product ☐ Exploded drawing

4 Key terms: write the letter of the matching description next to each term.

1. **Sketch model**	____	**A** Solid, non-working model showing shape and form; often used to test ergonomics.
2. **Block model**	____	**B** A model that looks and/or works like the final product, used for final testing before production.
3. **Scale model**	____	**C** Easy to change, realistic, can produce drawings and link to CAM/3D printing.
4. **Computer-generated model benefit**	____	**D** Model at a different scale for objects too large or small to model full size.
5. **Prototype**	____	**E** Quick, cheap model from scrap materials to generate and explore ideas.

5 Explain what is meant by ‘Simulation’. (2 marks)

6 Describe one drawback of rapid prototyping. (2 marks)

Check and reflect
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?